

■ Stochastic Modeling (622) – Course Overview

Welcome to Stochastic Modeling!

In this course, students increase their knowledge of modeling stochastic processes. Students will investigate advanced volatility models that upgrade Black-Scholes parameters to variables, increasing their stochastic modeling skills to address heteroskedasticity and variable costs as well as jump diffusions. Students will dive into discrete mathematics and Markov processes, including hidden Markov processes and Markov decision processes to financial applications, and will build a mathematical foundation for deep learning, a tool they will use for machine learning. Overall, students will be able to evaluate the assumptions, benefits, and difficulties associated with stochastic models.

■ Study Roadmap – Key Topics to Focus On

1. Foundations

- Review probability theory: distributions, expectation, variance, covariance.
- Understand conditional probability, independence, Bayes' theorem.

2. Stochastic Processes

- Markov chains (discrete and continuous time).
- Hidden Markov Models (HMMs).
- Markov Decision Processes (MDPs).

3. Volatility Models

- Understand Black–Scholes model assumptions.
- Study GARCH and other heteroskedasticity models.
- Explore stochastic volatility and jump diffusion models.

4. Poisson & Renewal Processes

- Poisson process properties and interarrival times.
- Applications in finance and risk modeling.

5. Brownian Motion & SDEs

- Properties of Wiener process.
- Ito's Lemma and applications.
- Geometric Brownian Motion (GBM) for stock prices.

6. Applications in Finance

- Option pricing: Black–Scholes, extensions.
- Risk-neutral pricing.
- Portfolio optimization under uncertainty.

7. Machine Learning Connection

- Mathematical foundation for deep learning.
- How stochastic processes inform ML algorithms.
- Randomness in optimization (SGD, dropout).

8. Practical Skills

- Use Python/R for stochastic simulations.
- Monte Carlo methods.
- Numerical solution of SDEs.

■ Stochastic Modeling (622) Cheat Sheet

1. Random Variables & Distributions

Bernoulli, Binomial, Poisson, Normal, Exponential
Mean (μ), Variance (σ^2), Standard Deviation (σ)
PDF vs CDF

2. Conditional Probability & Bayes' Theorem

$$P(A|B) = P(A \cap B) / P(B)$$
$$P(A|B) = [P(B|A)P(A)] / P(B)$$

3. Expectation & Variance

$$E[X] = \sum x P(X=x) \text{ (discrete)}, E[X] = \int x f(x) dx \text{ (continuous)}$$
$$\text{Var}(X) = E[X^2] - (E[X])^2$$

4. Covariance & Correlation

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$
$$\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$$

5. Law of Large Numbers & Central Limit Theorem

LLN: Sample average $\rightarrow$ expected value as $n \rightarrow \infty$
CLT: $\bar{X}_n \approx N(\mu, \sigma^2/n)$ for large n

6. Markov Chains

Transition matrix $P = [p_{ij}]$
Chapman–Kolmogorov: $P^{(n+m)} = P^n P^m$
Stationary distribution π : $\pi P = \pi$

7. Poisson Process

$N(t)$: number of arrivals by time t
Stationary & independent increments
 $P(N(t)=k) = (\lambda t)^k e^{-(\lambda t)} / k!$

8. Brownian Motion (Wiener Process)

$W(0)=0$, continuous paths
Independent, normally distributed increments
 $W(t) - W(s) \sim N(0, t-s)$

9. Ito's Lemma (SDEs)

If $dX = \mu(X,t)dt + \sigma(X,t)dW$, $Y=f(X,t)$:

$$dY = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW$$

10. Applications in Finance

Geometric Brownian Motion: $dS = \mu S dt + \sigma S dW$

Black–Scholes (Call Option): $C = S N(d) - K e^{-rt} N(d)$